

Identifying errors in proofs - Set 1 - Solutions

Specification Err1, Err2.

Answer key

Question	Answer	Topic
1	A	Identifying errors in proofs
2	A	Identifying errors in proofs
3	A	Identifying errors in proofs
4	A	Identifying errors in proofs
5	A	Identifying errors in proofs
6	A	Identifying errors in proofs
7	A	Identifying errors in proofs
8	C	Identifying errors in proofs
9	A	Identifying errors in proofs
10	A	Identifying errors in proofs
11	A	Identifying errors in proofs
12	A	Identifying errors in proofs

Worked solutions

Question 1 ERR-01 · Identifying errors in proofs · spec Err2

A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.

II So $x^2 - 7x + 6 = 0$.

III So $(x-1)(x-6) = 0$.

IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

Answer: A

Squaring can introduce extraneous solutions, so line IV should check them: $x = 1$ gives $\sqrt{4} = 2 \neq -2$. Only $x = 6$ works. Lines I-III are valid deductions.

Question 2 ERR-02 · Identifying errors in proofs · spec Err2

Here is a purported proof that $1 = 2$.

I Let $a = b$, with $a \neq 0$.

II Then $a^2 = ab$.

III So $a^2 - b^2 = ab - b^2$, i.e. $(a-b)(a+b) = b(a-b)$.

IV Dividing both sides by $(a-b)$: $a+b = b$.

V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

Answer: A

Since $a = b$, $a - b = 0$ and dividing by zero is invalid.

Question 3 ERR-03 · Identifying errors in proofs · spec Err2

A student tries to solve $\sin \theta = \sin 40^\circ$ for $0^\circ \leq \theta < 360^\circ$.

I Since $\sin \theta = \sin 40^\circ$, we have $\theta = 40^\circ$.

II The sine function has period 360° , so $\theta = 40^\circ + 360^\circ k$.

III In the given interval the only solution is $\theta = 40^\circ$.

Which of the following is correct?

Answer: A

'If $\sin \alpha = \sin \beta$ then $\alpha = \beta$ ' is a classic invalid deduction; $\sin(180^\circ - 40^\circ) = \sin 40^\circ$ too.

Question 4 ERR-04 · Identifying errors in proofs · spec Err1

Claim: for all real x , if $x^2 > 9$ then $x > 3$.

- I Suppose $x^2 > 9$.
- II Taking square roots of both sides, $x > 3$.
- III This proves the claim.

Which of the following is true?

Answer: A

$x^2 > 9$ gives $|x| > 3$, i.e. $x > 3$ or $x < -3$. The claim itself is false, and line II is where the invalid deduction happens.

Question 5 ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

- I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.
- II $(x + 1)^2 \geq 0$ for all real x .
- III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .
- IV Hence the equation has no real solutions.

Which of the following is correct?

Answer: A

Every step is valid.

Question 6 ERR-10 · Identifying errors in proofs · spec Err2

Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

- I Suppose $a > b$.
- II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.
- III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

Answer: A

$a = 1, b = -1$: $a > b$ but $\frac{1}{a} = 1 > -1 = \frac{1}{b}$. Dividing by ab without knowing its sign is invalid (and $b = 0$ would be division by zero).

Question 7 ERR-12 · Identifying errors in proofs · spec Err2

A student solves $x^2 = 4x$.

- I Divide both sides by x : $x = 4$.
- II So the only solution is $x = 4$.

Which of the following is correct?

Answer: A

Factorise instead: $x(x - 4) = 0$ gives $x = 0$ or $x = 4$.

Question 8 ERR-14 · Identifying errors in proofs · spec Err2

Claim: if x and y are real with $x^2 = y^2$, then $x = y$.

Which of the following is a counterexample to the claim?

Answer: C

A counterexample must satisfy the hypothesis ($x^2 = y^2$) and fail the conclusion ($x \neq y$).

Question 9 ERR-16 · Identifying errors in proofs · spec Err2

A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?

Answer: A

A number divisible by 2 is not prime unless it equals 2. The conclusion 'every prime is odd' is false.

Question 10 ERR-18 · Identifying errors in proofs · spec Err2

A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

Answer: A

Dividing by $\cos x$ loses $\cos x = 0$, i.e. $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$: two solutions.

Question 11 X2-25 · Identifying errors in proofs · spec Err2

A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3)$.

II So $(x - 2)(x - 3) > 0$.

III So $x - 2 > 0$ and $x - 3 > 0$.

IV So $x > 3$.

Which of the following is correct?

Answer: A

The full solution set is $x < 2$ or $x > 3$.

Question 12 ERR-05 · Identifying errors in proofs · spec Err2

A student solves the inequality $\frac{x+1}{x-2} < 3$.

I Multiply both sides by $(x-2)$: $x+1 < 3x-6$.

II So $7 < 2x$.

III So $x > \frac{7}{2}$.

Which of the following is correct?

Answer: A

For $x < 2$ the multiplier is negative and the inequality reverses; indeed $x = 0$ gives $-\frac{1}{2} < 3$, a solution missed by the student. The full solution set is $x < 2$ or $x > \frac{7}{2}$.